

Experiment G — Quadrature Speed/Equivariance Pareto

What is the right way to compare DH vs GL? **Same equivariance error**, then ask how much wall-clock time each method needs to reach it.

Setup (fairchem default EquiformerV2, lmax=6, mmax=2)

Real EquiformerV2Backbone from fairchem: 12 layers, 128 channels, 24 S2Activation modules. NVIDIA A100-SXM4-40GB, idle.

These are the function-signature defaults of EquiformerV2 in fairchem (`num_layers=12`, `sphere_channels=128`, `lmax_list=[6]`, `mmax_list=[2]`); the published OC20 SOTA model is larger (20 layers, `lmax=8`, `mmax=8`, $\sim 150\text{M}$ params), so the +31% savings here is a lower bound for the production model. The 24 modules = $2 \times \text{num_layers}$ (one S2 act in graph attention + one in FFN per TransBlockV2).

Config	Method	Grid (lat $\times$ lon)	Points
DH default	e3nn equiangular	14×5	70
DH 2 $\times$	DH with <code>resolution=4(L+1)</code>	28×28	784
GL min	custom GL	7×13	91
GL match-DH	custom GL	14×13	182
GL 2 $\times$	custom GL	14×28	392

Two measurements at the same (lmax, mmax)

Equivariance error of one S2-Act kernel (random coefficients, 5 rotations $\times$ 10 inputs); SiLU activation. *Source:* `verify_gl_grid.py`.

Wall-clock for one full forward (QM9, batch=8), 90 measurements per config (3 runs $\times$ 30 iter), `cuda.sync()` around each, fresh model per config. *Source:* `verify_speedup_fairchem_default.json`.

Combined results

Config	Pts	Equiv err	Fwd (ms, 95% CI)
DH default	70	4.43×10^{-1}	111.83 ± 0.79
DH 2$\times$	784	3.27×10^{-1}	162.73 ± 1.23 (+45.5%)
GL min	91	4.29×10^{-1}	~ 111 (matches DH default)
GL match-DH	182	3.28×10^{-1}	112.23 ± 0.75 (+0.4% n.s.)
GL 2$\times$	392	3.28×10^{-1}	111.34 ± 0.76 (-0.4% n.s.)

Matched-equivariance comparison

For target equiv err = 3.28×10^{-1} (the saturation floor at this mmax cropping):

- Cheapest DH: **DH 2 $\times$ at 162.73 ms** (DH default at 70 pts only reaches 4.43×10^{-1} , not enough)
- Cheapest GL: **GL match-DH at 112.23 ms**
- **Savings: 50.5 ms** = $(162.73 - 112.23)/162.73 = 31.0\%$ **time reduction**

So: at matched equivariance accuracy, GL saves $\sim 31\%$ wall-clock per forward.

Equivalently: GL achieves DH 2×’s equivariance at the same wall-clock as DH default; the +50.5 ms premium goes away.

Caveats

1. Result is for this specific (lmax=6, mmax=2) config and this GPU. Different model depths or hardware will scale it.
2. Equivariance is measured on *one* S2-Act kernel with random inputs, not on a trained model’s intermediate features. Trained-model features are smoother (absolute equiv numbers will be much lower); the *ranking* between quadratures should be preserved.
3. The kernel-level equivariance gap between GL match-DH (3.28×10^{-1}) and DH 2× (3.27×10^{-1}) is within rounding — they really are the same accuracy.
4. The mmax=2 cropping caps achievable equivariance at ~ 0.328 regardless of grid; lower error needs mmax=lmax (architectural change). At lmax=mmax both methods can reach $\sim 4 \times 10^{-3}$ floor; there GL needs roughly half the latitude points of DH (see `verify_gl_grid.py`).
5. We did not measure end-to-end equivariance on a fully trained fairchem default model. The matched-accuracy speedup claim rests on the kernel-level Pareto.

Earlier vs revised framing

The previous report claimed “DH 2× costs +45.5% over DH default”, which was correct but compared different equivariance levels. The fairer claim:

Switching DH 2× → GL match-DH saves **31% wall-clock per forward** while preserving the same equivariance error. Switching DH default → GL match-DH is **free** (same wall-clock) and substantially improves equivariance ($4.43 \times 10^{-1} \rightarrow 3.28 \times 10^{-1}$).

Reproduce

```
module load pytorch/2.6.0-1
# Equivariance for all configs at lmax=6, mmax=2
python3 experiments/verify_gl_grid.py
# Wall-clock: rigorous 3-run verification
PYTHON_CUDA_ALLOC_CONF=expandable_segments:True \
python3 experiments/expG_verify_speedup.py \
  --config fairchem_default --batch_size 8 \
  --n_repeats 3 --n_iter 30 --n_warmup 20
```